SentryScan: Assignment 3.1 — Multi-Agent Coordination, Negotiation & LangGraph

Course: 21CSE525T — Agent Technology
Assignment: 3.1 — Multi-Agent Coordination & Negotiation
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Repository: https://github.com/drgeekg/SentryScan
Target Branch: `assignment/3.1-multi-agent-coordination`
Git Tag: `assignment-3.1`
Status: Completed & Verified

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1. Executive Summary & Theoretical Context

Assignment 3.1 implements a distributed **Multi-Agent Coordination & Negotiation architecture** orchestrated by a compiled LangGraph state workflow, modeling the classical **FIPA Contract-Net Protocol (CNP)** (Smith, 1980; Wooldridge, Ch. 7).

1.1 Conflicting Agent Roles & Rational Self-Interest

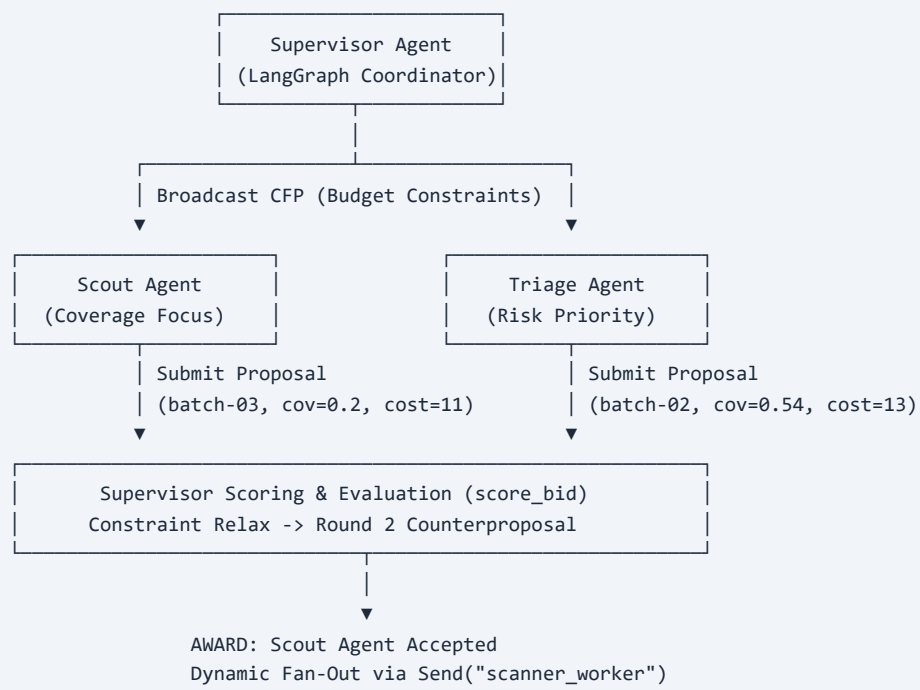
Rather than homogeneous workers, SentryScan coordinates specialized agents with distinct operational preferences: - **Scout Agent:** Maximizes *host coverage breadth* across the network fleet. - **Triage Agent:** Maximizes *critical severity risk mitigation*, prioritizing known high-risk hosts. - **Supervisor Agent:** Acts as the auctioneer, balancing budget constraints and optimizing global system utility.

1.2 Bounded Contract-Net Negotiation Protocol

1. Supervisor broadcasts `Call for Proposals (CFP)` specifying constraints (max cost, deadline).
2. Scout and Triage compute candidate plans and return formal `PROPOSE` bids.
3. Supervisor evaluates bids via an objective utility function. If constraints are unsatisfied, Supervisor issues `COUNTERPROPOSE` with relaxed tolerances.
4. When a bid satisfies criteria, Supervisor issues `ACCEPT` to the winning contractor and `REJECT` to the other. Winning agent sends `CONFIRM`.
5. Negotiation terminates within a hard bounded limit (≤ 5 rounds); if consensus is unachievable, terminates safely in `UNRESOLVED`.

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2. Architecture & Implementation Specification



2.1 Dynamic Runtime Fan-Out via LangGraph `Send`

Once the winning proposal is awarded, the supervisor graph dynamically branches execution to parallel worker instances:

```
# Dispatches parallel scanner workers per awarded telemetry batch
return [Send("scanner_worker", {"batch_id": b, "run_id": state["run_id"]})
        for b in awarded_proposal.batch_ids]
```

3. Rubric Compliance & Verification Matrix

Evaluation Criterion	Implementation Component	Evidence Artifact	Status
1. LangGraph Supervisor Workflow	`src/sentryscan/graphs/supervisor.py`: `StateGraph(SupervisorState)` with negotiation, worker fan-out, and result aggregation	`tests/unit/test_supervisor_graph.py` `demos/demo_assignment_3_1.py` Step 2	**PASSED**
2. Role Separation (Scout vs. Triage)	`src/sentryscan/agents/scout.py` (coverage-maximizing) and `src/sentryscan/agents/triage.py` (risk-prioritizing)	`tests/unit/test_proposal_roles.py` `evidence/negotiation_timeline.md`	**PASSED**
3. Contract-Net Negotiation Protocol	`src/sentryscan/negotiation/protocol.py`: `CFP` $ o$ `PROPOSE` $ o$ `COUNTERPROPOSE` $ o$ `ACCEPT` / `REJECT` $ o$ `CONFIRM`	`evidence/negotiation_timeline.md` `tests/unit/test_negotiation.py`	**PASSED**
4. Inline Bid Scoring Seam (`score_bid`)	`src/sentryscan/negotiation/scoring.py`: Objective formula over normalized coverage, cost, time, and risk with hard constraint filtering ($-\infty$)	`tests/unit/test_negotiation.py` (`test_score_bid_valid`, `test_score_bid_constraint_violations`)	**PASSED**
5. Dynamic Runtime Fan-Out via `Send`	`src/sentryscan/graphs/supervisor.py`: Returns `[Send("scanner_worker", ...)]` dynamically dispatching parallel worker instances per awarded batch	`tests/unit/test_supervisor_graph.py` `evidence/run_log.txt` Step 2	**PASSED**
6. Provider-Neutral Advisory LLM Layer	`src/sentryscan/llm/`: `LLMProvider` protocol with `DeterministicLLMProvider` (offline template generation) and `FakeLLMProvider`	`tests/unit/test_llm.py` `tests/integration/test_multi_agent_coordination.py`	**PASSED**
7. Bounded Loop & `UNRESOLVED` Termination	Hard limit (default: 5 rounds) terminating safely in `UNRESOLVED` when constraints cannot be satisfied	`tests/unit/test_negotiation.py` (`test_negotiation_unresolved_after_max_rounds`)	**PASSED**
8. Determinism Contract (LLM Off)	Deterministic single-threaded execution ensures 0 drift across runs with LLM off or deterministic provider	`tests/integration/test_multi_agent_coordination.py` (`test_trace_identical_with_llm_off`)	**PASSED**
9. Trace Persistence & Single-Writer Auditing	Sequential JSONL event log capturing every message, proposal, and finding via `src/sentryscan/observability/recorder.py`	`evidence/trace_excerpt.jsonl`	**PASSED**

4. Execution Results & Empirical Evidence

4.1 Turn-by-Turn Negotiation Timeline (`evidence/negotiation_timeline.md`)

Step	Sender	Receiver	Performative	Summary	Nonce
1	`supervisor`	`scout`	`CFP`	Round 1 Constraints: `max_cost=10.5, max_time=250.0`	`30877432d1026706`
2	`supervisor`	`triage`	`CFP`	Round 1 Constraints: `max_cost=10.5, max_time=250.0`	`d7e805da846a32c3`
3	`scout`	`supervisor`	`PROPOSE`	Batches: `['batch-03']`, Cost: 11.0, Coverage: 0.20	`bb81e3c29b621792`
4	`triage`	`supervisor`	`PROPOSE`	Batches: `['batch-02']`, Cost: 13.0, Coverage: 0.54	`73c8eb5bb682575e`
5-6	`supervisor`	Both	`COUNTERPROPOSE`	Round 2 Constraints Relaxed: `max_cost=12.075, max_time=287.5`	—
7-8	Both	`supervisor`	`PROPOSE`	Re-submitted matching proposals under relaxed budget	—
9	`supervisor`	`scout`	`ACCEPT`	**AWARDED:** `['batch-03']`, Score: `-0.1256`	`71722f244f58d669`
10	`supervisor`	`triage`	`REJECT`	Rejected: Lower relative utility score under cost cap	`cbee3772a0770217`
11	`scout`	`supervisor`	`CONFIRM`	Award confirmed, worker execution begins	`21a278f64f7fd633`

5. Submission Package & Reproduction Instructions

5.1 Exact Submission Components

1. **Report Document:** `assignments/assignment_3_1/deliverables/SentryScan_Assignment_3_1_Report.pdf`
2. **Repository URL:** `https://github.com/drgeekg/SentryScan`
3. **Target Branch:** `assignment/3.1-multi-agent-coordination`
4. **Immutable Git Tag:** `assignment-3.1`
5. **Packaged ZIP Archive:** `dist/SentryScan_assignment_3_1.zip`
6. **SHA-256 Checksum:** `dist/SentryScan_assignment_3_1.zip.sha256`

5.2 Verification Commands

```
# Run multi-agent negotiation demonstration
python demos/demo_assignment_3_1.py

# Execute supervisor graph and negotiation tests
pytest -v tests/unit/test_supervisor_graph.py tests/unit/test_negotiation.py
```